Digital Twins for Human-Centered Support and Collaborative Robotics

Cédric Simar · 16/03/2026 · MLG

About myself

Cédric Simar

Oct. 2026: Postdoctoral Researcher **WEL-T** at **ULB**

Last-year PhD Student at **ULB** (MLG)

- Electroencephalography EDT
- Myoelectric robotic hand control

Lecturer at **UMONS** (Neuroscience dep.)

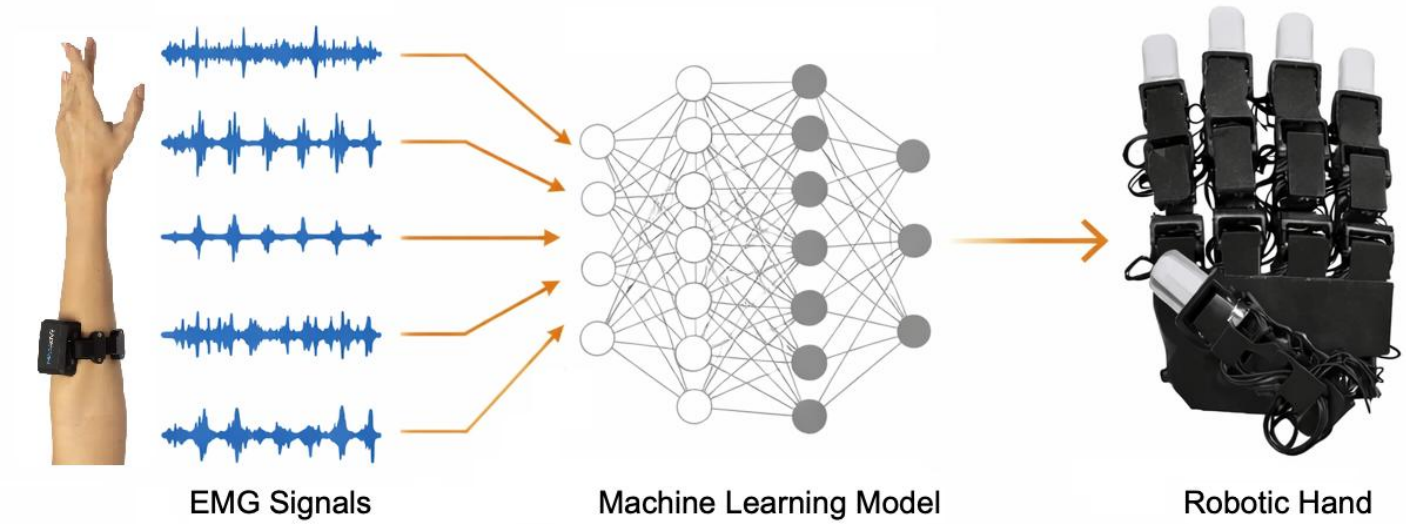
- Introduction to AI and Brain-Computer Interfaces

MSc. in Computer Science at **ULB** (MLG)

Current work

Myoelectric robotic hand control

Objective: develop novel machine learning models and frameworks to fully control all degrees of freedom of a robotic hand using myoelectric signals.



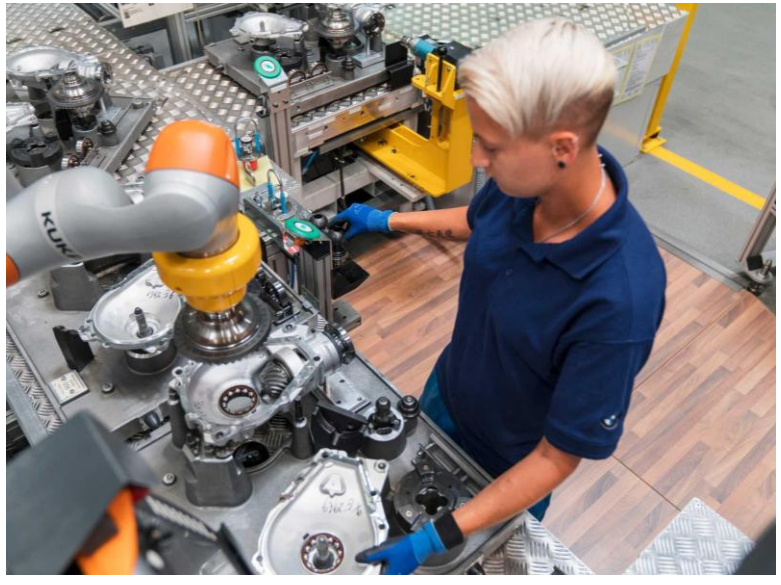
WEL-T Use Case 1: Collaborative Robotics

The **physical robotic arm** assists workers at a workstation in performing high-precision or tedious tasks, simultaneously collecting data on workstation states and human fatigue.

Its **digital twin** runs what-if scenarios in real-time to determine optimal strategies for human-centered assistance and enhancing human capabilities.

Feasibility assessment: **high**, supported by clear KPIs, established Isaac Sim platform, expertise in biosignal analysis.

Societal impact: reduces worker's muscular effort, prevents injuries, while improving productivity. Collaborative systems can later be extended to people with disabilities.



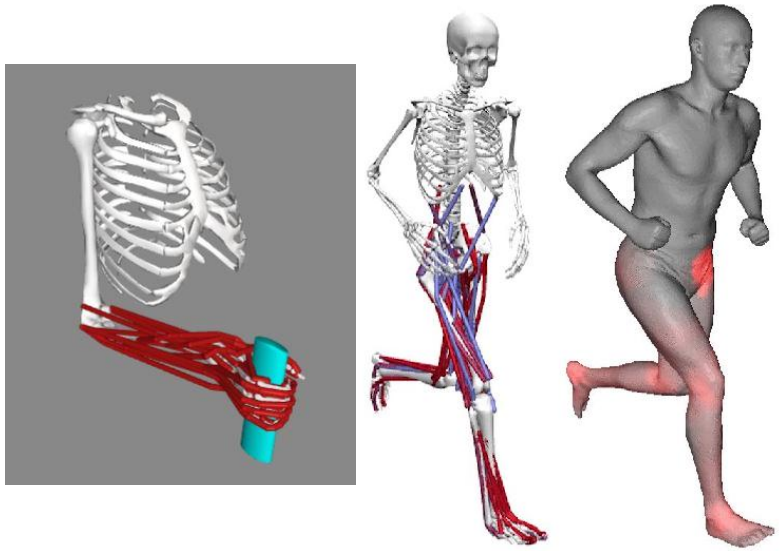
WEL-T Use Case 2: Rehabilitation

The patient performs rehabilitation exercises while EMG and motion tracking data are recorded.

The **digital twin** is a subject-specific musculoskeletal model that runs what-if scenarios to help therapists identifying the exercises that minimize risks of injuries or harmful patterns.

Feasibility assessment: **medium**, depend on medical and sport cooperations, existing models but no platform, expertise in pro athletes biosignal analysis (David Goffin, J.H. academy, J. Borlée)

Societal impact: improve rehabilitation effectiveness and safety. The digital twin can later be extended to professional athletes for early explainable detection and prevention of injury risks.



12-Month Research Plan

Months 1-6
Apr-Sep 2026

Months 7-9
Oct-Dec 2026

Months 10-12
Jan-Mar 2027

Discovery, selection, planning

- Gather stakeholder input and run discovery meetings
- Assess appetite and technical feasibility for both use cases
- Select the priority use case
- Define work packages, milestones, and deliverables
- Assess key risks, expected impact on deliverables, and mitigations

Choose use case

Collaborative Robotics path

Building foundations

- Source robotic components
- Build the interface framework and core foundations
- Bench-test haptic sensing options
- Assemble the robotic hand

Integrating and extending

- Assemble the robotic arm
- ROS 2 integration and synchronization
- Extend and integrate the digital twin stack for robotics

Rehabilitation path

Building the framework

- Source biosignal acquisition hardware
- Build the interface framework and link it to anatomical models

Validating and integrating

- Synchronize and validate biosignal devices
- Validate the framework with domain experts
- Integrate the framework into the digital twin stack